

Wireless Architectures (Part-2)

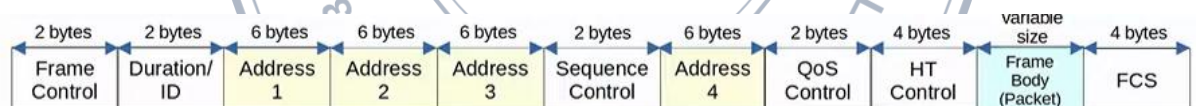
- Topics covered:
 - 802.11 messages/frame format
 - Autonomous APs
 - Lightweight APs
 - Cloud-based APs
 - Wireless LAN Controller (WLC) Deployments

802.11 Frame Format

802.11 Frame Format

802.11 frames are more complex than Ethernet frames. They contain more header fields, and some fields may or may not be present depending on the 802.11 version and message type. For example, up to four address fields can exist, but not every frame uses all four.

Frame Structure



Frame Control

Length: 2 bytes (16 bits)

Provides information such as the message type and subtype.

Duration/ID

Length: 2 bytes

Depending on the message type:

- Indicates the transmission duration in microseconds.
- Or acts as an identifier for the association between a wireless client and an AP.

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Address Fields

Up to four addresses can be present:

1. Destination Address (DA): Final recipient of the frame.
2. Source Address (SA): Original sender.
3. Receiver Address (RA): Immediate recipient, not necessarily the final destination.
4. Transmitter Address (TA): Immediate sender, not necessarily the original source.

Wireless networks require these additional address fields because of their unique operational requirements.

Sequence Control

Used to reassemble fragments.

Eliminates duplicate frames.

QoS Control

Used for Quality of Service (QoS).

Prioritizes specific traffic types.

HT (High Throughput) Control

Added in 802.11n.

Enables high-throughput operations.

802.11n (Wi-Fi 4) is also called *High Throughput Wi-Fi*.

802.11ac (Wi-Fi 5) is also called *Very High Throughput (VHT) Wi-Fi*.

Frame Body

Contains the packet encapsulated inside the 802.11 frame.

FCS (Frame Check Sequence)

Located in the trailer.

Used for error checking, similar to Ethernet.

Key Point

Not every 802.11 frame contains all of these fields. The exact format depends on the 802.11 version and message type being used.

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802.11 Association Process

A wireless station must be associated with an access point before it can send traffic through it. APs bridge traffic between wireless stations and other devices, including devices on the wired network.

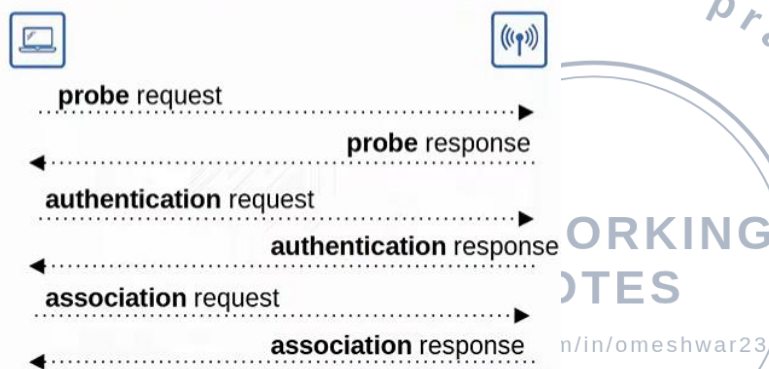
Connection States

There are three connection states:

1. Not authenticated and not associated
2. Authenticated but not associated
3. Authenticated and associated

Only the third state allows the station to send traffic through the AP.

Association Sequence



Step 1: Discovery

The station sends a **probe request** to discover available APs and BSSs. The AP replies with a **probe response** indicating its availability.

Step 2: Authentication

The station and AP perform an authentication exchange. If successful, the station becomes **authenticated but not yet associated**.

Step 3: Association

The station sends an association request and receives an association response. If successful, the station becomes **authenticated and associated**, allowing it to send traffic through the AP.

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Scanning Methods

A station can discover a BSS in two ways:

Active Scanning:

- Station sends probe requests.
- Station listens for probe responses from APs.

Passive Scanning

- Station listens for beacon messages.
- APs periodically send beacon messages to advertise the BSS.

ASCII Diagram

Active Scanning

Station -----> Probe Request

Station <----- Probe Response

Passive Scanning

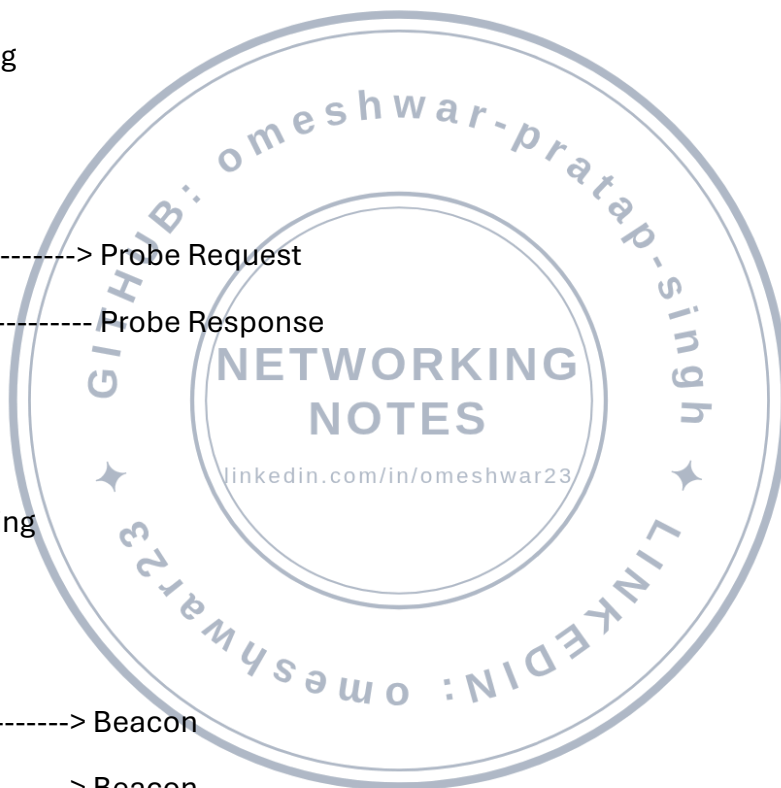
AP -----> Beacon

AP -----> Beacon

AP -----> Beacon

Station listens for beacons

Using either active or passive scanning, the station learns about the AP and BSS before authentication and association occur.



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802.11 Message Types

There are three message types in 802.11.

1. Management Frames: Used to manage the BSS.

Examples:

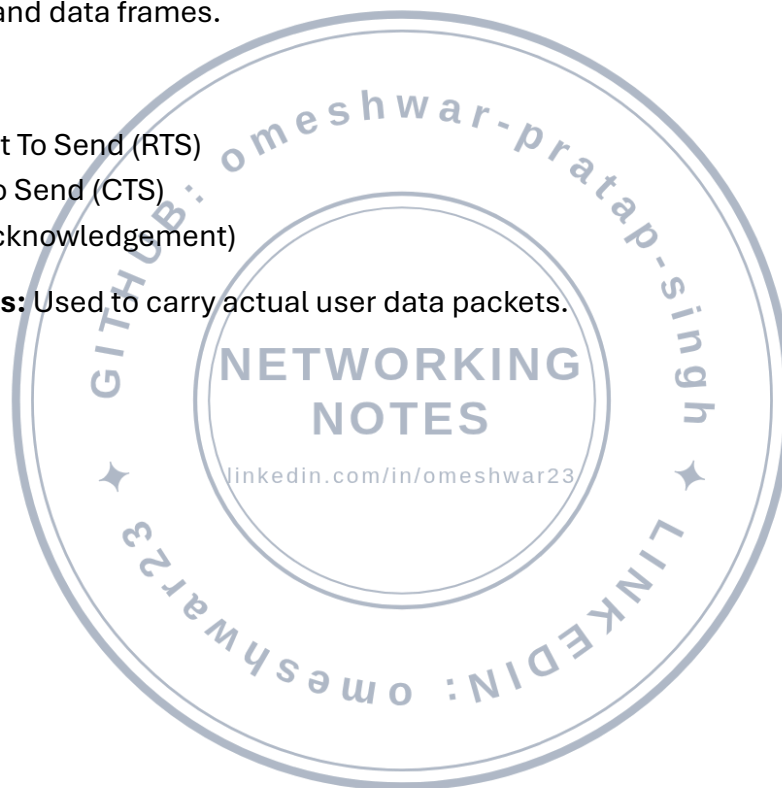
1. Beacon
2. Probe
3. Authentication
4. Association

2. Control Frames: Used to control access to the radio medium and assist in delivering management and data frames.

Examples:

1. Request To Send (RTS)
2. Clear To Send (CTS)
3. ACK (Acknowledgement)

3. Data Frames: Used to carry actual user data packets.



Autonomous Access Points

Autonomous APs

There are three main AP deployment methods:

1. Autonomous
2. Lightweight
3. Cloud-Based

Autonomous APs are self-contained systems that do not depend on a Wireless LAN Controller (WLC). They operate independently and are configured individually.

Configuration Methods

Autonomous APs can be configured through: Console cable, Telnet, SSH, HTTP, HTTPS Web GUI

For small networks this is acceptable, but individually configuring APs becomes unrealistic in large networks.

Autonomous AP Characteristics

Management:

Each AP requires its own management IP address.

Remote management uses Telnet, SSH, HTTP, or HTTPS.

Radio Configuration:

Each AP must be configured manually for:

- Transmit power
- Radio channel selection

Security: Security policies such as ACLs are configured individually on each AP.

QoS: QoS settings are configured separately on every AP.

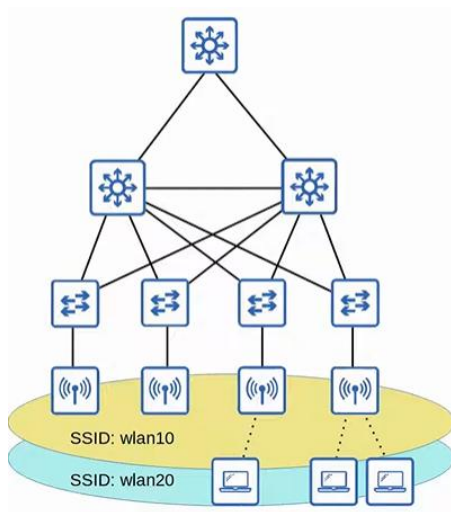
Monitoring:

No centralized monitoring or management exists.

Each AP functions as an independent system.

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Autonomous AP Network Design

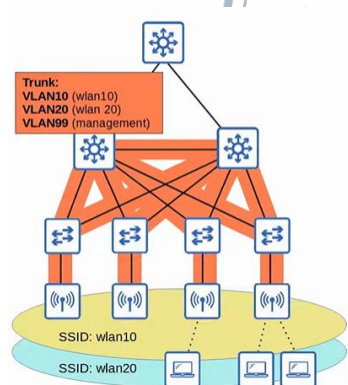


This design uses multiple APs providing two SSIDs: wlan10 and wlan20.

Trunk Requirement

Each autonomous AP should connect to the wired network using a trunk link.

VLANs Carried



Even when only one SSID is provided, a trunk is recommended because management traffic should remain separate from normal user traffic.

Management traffic should be placed in a separate subnet and VLAN from regular data traffic.

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Traffic Flow

With autonomous APs, traffic takes a direct path.

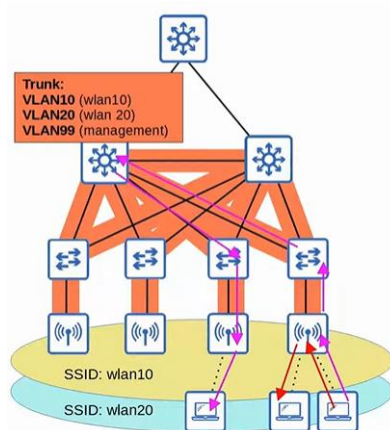
Between Different APs:

PC ---> AP ---> Switches ---> AP ---> PC

Between Clients on the Same AP:

Client ---> AP ---> Client

Traffic does not need to travel through the wired network when both clients are associated with the same AP.



VLAN Stretching Issues

In this design, VLANs extend across the entire network. This is considered a bad practice.

Reason 1: Large Broadcast Domains:

Each VLAN is a broadcast domain.

Broadcast traffic is flooded throughout the entire network.

Reason 2: Spanning Tree Disables Links:

Spanning Tree blocks some links.

Disabled links reduce available bandwidth.

Reason 3: VLAN Administration:

Adding or removing VLANs becomes labor-intensive.

Changes must be made across many switches.

Lightweight APs

Lightweight APs (Split-MAC Architecture)

The next major AP deployment model is the **lightweight AP**. In this architecture, AP functions are divided between the lightweight AP and a Wireless LAN Controller (WLC).

(Function Distribution)

Lightweight AP Responsibilities

Lightweight APs perform real-time wireless operations such as:

1. Transmitting and receiving RF traffic
2. Encryption and decryption of traffic
3. Sending beacon messages
4. Sending probe messages

WLC Responsibilities

The WLC centrally performs functions such as:

1. RF management
2. Security management
3. QoS management
4. Client authentication
5. Client association
6. Roaming management

The WLC controls these functions for many lightweight APs simultaneously. This architecture is called **Split-MAC (Split Media Access Control)** because AP functions are divided between the AP and the WLC.

The WLC is also used to centrally configure lightweight APs. Instead of logging into each AP individually and configuring them one by one, all configuration is performed from the WLC.

The WLC may be located:

1. In the same subnet and VLAN as the APs
2. In a different subnet and VLAN

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AP and WLC Authentication

The WLC and lightweight APs authenticate each other using **digital certificates** installed on both devices.

These certificates follow the **X.509 standard**, the same standard used by websites to prove authenticity on the Internet.

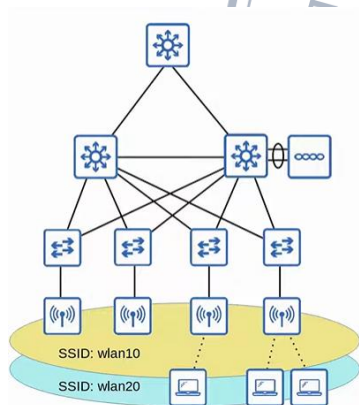
This ensures that only authorized APs can join the network.

CAPWAP Communication

Lightweight APs communicate with the WLC using **CAPWAP** (Control and Provisioning of Wireless Access Points).

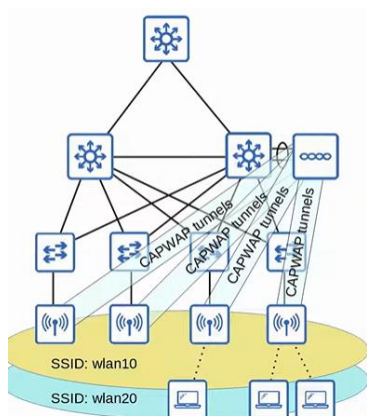
CAPWAP is based on an older protocol called **LWAPP** (Lightweight Access Point Protocol).

Lightweight AP Network



CAPWAP Tunnels

Two CAPWAP tunnels are created between each AP and the WLC.



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Each AP establishes:

1. Control Tunnel
2. Data Tunnel

CAPWAP Control Tunnel

Characteristics:

UDP Port: 5246

Used for:

1. AP configuration
2. AP control
3. AP management

Security: All traffic in the control tunnel is encrypted by default.

CAPWAP Data Tunnel

Characteristics:

UDP Port: 5247

Carries wireless client traffic

All wireless client traffic is sent through this tunnel to the WLC.

Traffic:

1. Does not go directly to the wired network.
2. Does not go directly to another client associated with the same AP.

It must first pass through the WLC.

Security:

Data tunnel traffic is not encrypted by default.

It can be encrypted using **DTLS (Datagram Transport Layer Security)**.

DTLS is similar to TLS, except DTLS uses UDP while regular TLS uses TCP.

Switch Port Requirements

This is another major difference between lightweight APs and autonomous APs.

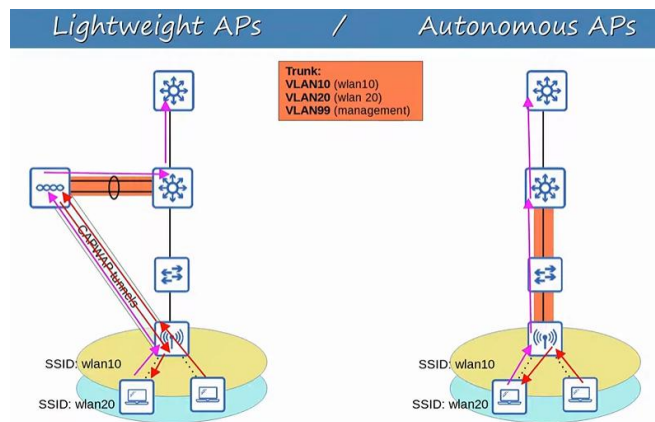
Lightweight AP: Connects to a switch access port and, Does not require a trunk link between the AP and switch.

WLC: Requires a trunk link to the wired network and, The WLC performs the VLAN mapping role instead of the AP.

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Traffic Flow Comparison

To understand the difference, compare lightweight APs (Split-MAC) and autonomous APs (Local-MAC).



Autonomous AP (Local-MAC)

Traffic to External Network: The AP forwards traffic directly into the wired network.

Traffic to Another Client on the Same AP: The traffic does not need to enter the wired network.

Lightweight AP (Split-MAC)

The AP does not form the boundary between wired and wireless networks.

The WLC performs that role.

Traffic to External Network:

Traffic is tunneled to the WLC first.

The WLC then forwards it into the wired network.

Traffic Between Clients on the Same AP:

Even traffic destined for another client associated with the same AP must first be tunneled to the WLC and then sent back.

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Benefits of Split-MAC Architecture

The Split-MAC architecture provides several advantages.

1. Scalability

Supports thousands of APs.

Much easier to build and maintain large wireless networks.

2. Dynamic Channel Assignment

The WLC can automatically select channels for APs.

Manual channel planning is not required.

3. Automatic Transmit Power Adjustment

The WLC can automatically configure transmit power levels.

This provides sufficient coverage while reducing interference.

4. Self-Healing Coverage

If an AP stops functioning, the WLC can increase the transmit power of nearby APs to help prevent coverage holes.

5. Seamless Roaming

Clients can move between APs without noticeable delay.

6. Client Load Balancing

If multiple APs are available, the WLC can associate a client with the least-used AP to balance load.

7. Security and QoS Management

Centralized management ensures consistent:

- Security policies
- QoS policies

across the network.

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Lightweight AP Modes

Just as autonomous APs can operate in different modes, lightweight APs also support multiple modes.

1. Local Mode

This is the default mode.

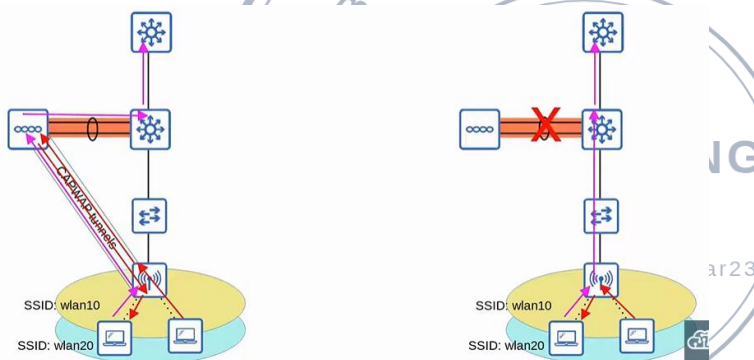
Characteristics

1. Provides one BSS or multiple BSSs.
2. Allows clients to associate with the AP.
3. Standard operational mode.

2. FlexConnect Mode

FlexConnect mode also provides one or more BSSs for clients.

In addition, it allows local switching of traffic if connectivity to the WLC is lost.



Normal Operation:

All traffic is tunneled to the WLC.

When WLC Connectivity Fails:

If CAPWAP tunnels to the WLC go down, the AP can locally forward traffic between the wired and wireless networks.

The AP behaves similarly to an autonomous AP and no longer needs to tunnel traffic to the WLC.

3. Sniffer Mode

In **Sniffer Mode**, the AP does **not** offer a BSS for clients.

Instead, the AP is dedicated to:

- Capturing 802.11 frames
- Sending captured frames to packet analysis software such as Wireshark

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The AP listens to wireless traffic and forwards the captured packets for analysis.

4. Monitor Mode

In **Monitor Mode**, the AP also does **not** offer a BSS for clients.

Instead, it is dedicated to:

- Receiving 802.11 frames
- Detecting rogue devices

If a rogue client is detected, the AP can send **de-authentication messages** to disassociate the rogue device from the AP.

5. Rogue Detector Mode

Rogue Detector Mode is similar to Monitor Mode, but differs in an important way.

In this mode:

- The AP does not use its radio.
- It only listens to traffic on the wired network.
- The WLC provides a list of suspected rogue AP and client MAC addresses.

The AP listens for ARP traffic on the wired network and correlates that information with data received from the WLC to detect rogue devices.

6. SE-Connect Mode (Spectrum Expert Connect)

SE-Connect stands for **Spectrum Expert Connect**.

In this mode:

- The AP does not offer a BSS.
- The AP is dedicated to RF spectrum analysis.
- Analysis is performed on all channels.

The AP can send RF spectrum information to software such as Cisco Spectrum Expert running on a PC.

This helps identify sources of interference.

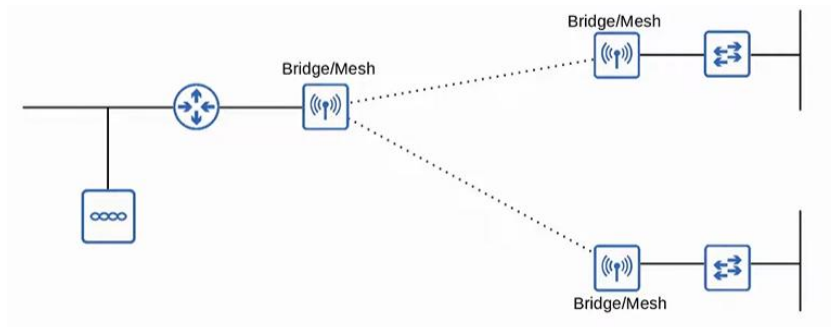
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7. Bridge / Mesh Mode

Like the autonomous AP's Outdoor Bridge mode, a lightweight AP can operate as a dedicated bridge between sites, including over long distances.

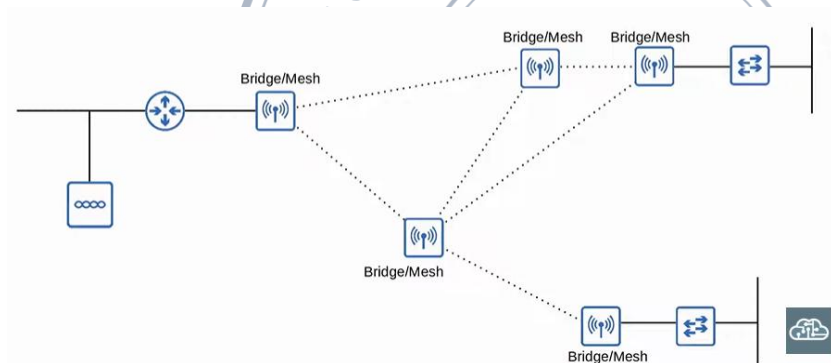
A mesh topology can also be created between APs.

Bridge Example:



This provides a dedicated wireless bridge between sites.

Mesh Example:



The APs form a mesh while connecting remote sites.

Cloud-based APs

Cloud-Based AP Architecture

The final AP architecture is *Cloud-Based AP Architecture*.

This architecture sits between:

1. Autonomous AP Architecture
2. Split-MAC Architecture

in terms of functionality.

The basic idea is: Autonomous APs that are centrally managed in the cloud.

A common example is **Cisco Meraki**.

Meraki Dashboard

The Meraki Dashboard is a web-based management interface.

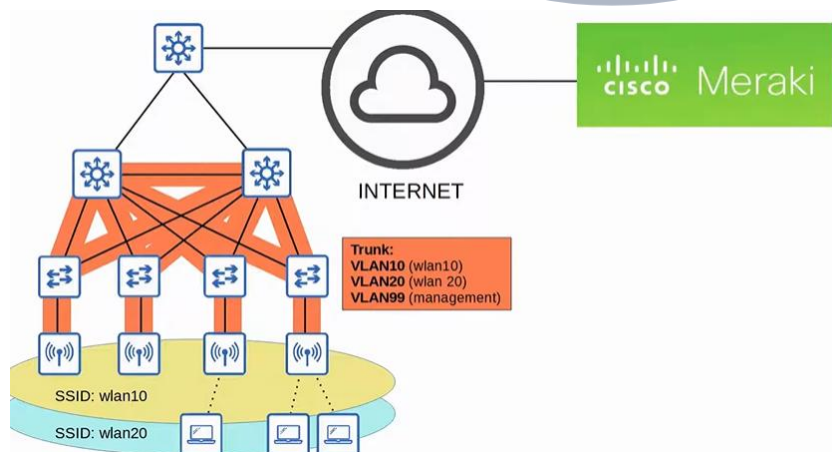
It can be used to:

- Configure APs
- Monitor the network
- Generate performance reports

Like a WLC, it can also determine:

- Which channel an AP should use
- Which transmit power an AP should use

Cloud-Based Architecture



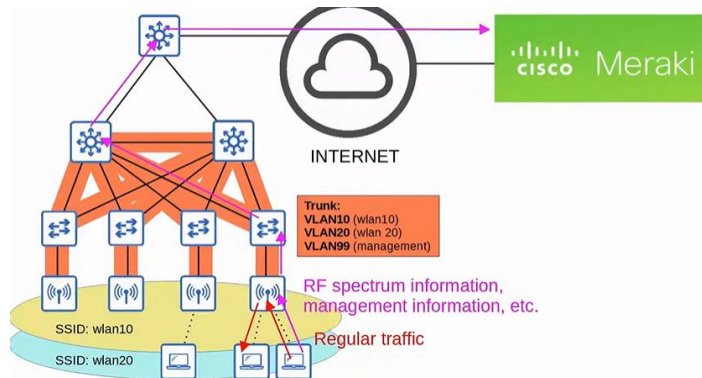
The APs are centrally managed through the cloud.

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Traffic Flow:

Only management/control traffic is sent to the cloud.

Regular user traffic is sent directly to the wired network.



Management traffic and user traffic remain separate.

Meraki Traffic Separation

Meraki Management Data

----->

Client --> AP --> Network --> Internet --> Meraki Cloud

User Data

----->

Client --> AP --> Network --> Web Server

File Server

Management information goes to the Meraki cloud.

User traffic goes directly to its destination.

Key Characteristics

Cloud-based AP architecture can be viewed as:

- Autonomous APs
- Centrally managed through the cloud

Unlike Split-MAC architecture:

- User traffic is not tunneled through a controller.
- Only management/control information is sent to the cloud.

Wireless LAN Controller (WLC) Deployments

WLC Deployment Models

The final topic is Wireless LAN Controller deployment.

These deployment models apply only to **Split-MAC Architecture**.

They do **not** apply to:

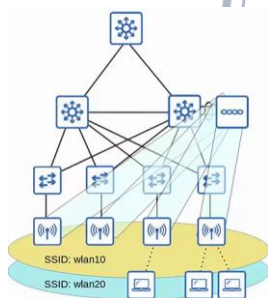
1. Autonomous AP Architecture
2. Cloud-Based AP Architecture

There are four main deployment models.

1. Unified WLC Deployment

A Unified WLC is:

- A dedicated hardware appliance
- Located centrally in the network



Characteristics:

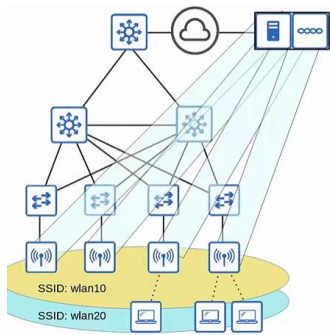
- Separate hardware device
- Centralized deployment
- Supports approximately **6000 APs**
- Additional WLCs can be added if needed
- Suitable for large enterprise campuses

2. Cloud-Based WLC Deployment

The WLC is:

- A virtual machine
- Running on a server
- Usually located in a private cloud/data center

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Characteristics:

- Supports approximately **3000 APs**
- Additional WLC VMs can be added if required

Important Note:

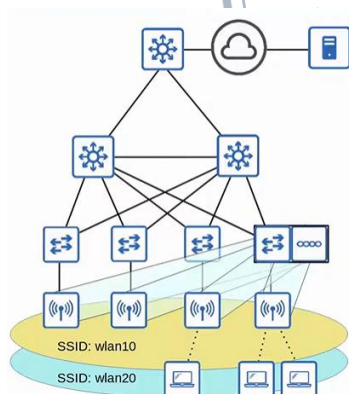
This is not the same as the cloud-based AP architecture.

Here:

- APs are still lightweight APs.
- Only the WLC is cloud-hosted.

3. Embedded WLC Deployment

The WLC is integrated into a switch.



Characteristics:

- WLC functionality built into a switch
- Supports approximately **200 APs**
- Additional embedded WLC switches are required to support more APs
- Suitable for smaller campuses

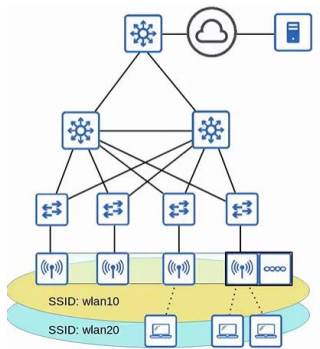
4. Cisco Mobility Express Deployment

The WLC functionality is integrated into an AP.

The AP containing the controller establishes internal CAPWAP tunnels.

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Other APs also connect to it.



Characteristics:

- WLC functionality resides in an AP
- Supports approximately **100 APs**
- Additional Mobility Express APs are required for larger deployments
- Suitable for small branch offices

WLC Deployment Summary

Deployment Type	WLC Location	Approximate AP Support	Typical Use
Unified WLC	Dedicated hardware appliance	~6000 APs	Large enterprise campus
Cloud-Based WLC	VM in private cloud/data center	~3000 APs	Large deployments
Embedded WLC	Integrated in switch	~200 APs	Smaller campuses
Cisco Mobility Express	Integrated in AP	~100 APs	Small branch offices